

Index No: 25000799

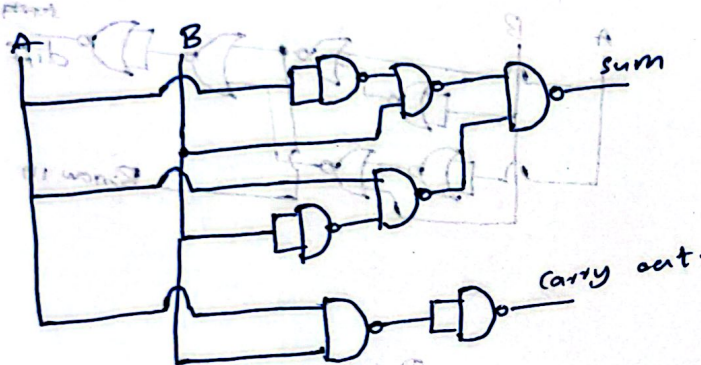
①

A	B	Sum	Carry out
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

$$\text{Sum} = \overline{A}B + A\overline{B}$$

$$\text{Carry out} = AB + \overline{A}\overline{B}$$

$$\begin{aligned} \overline{A}B + A\overline{B} &= \overline{\overline{\overline{A}B + A\overline{B}}} \\ &= \overline{\overline{A}B} \cdot \overline{A\overline{B}} \end{aligned}$$



②

A	B	C _{in}	Sum	Carry out
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

$$\text{Sum} = \overline{A}\overline{B}C_{in} + \overline{A}B\overline{C}_{in} + A\overline{B}\overline{C}_{in} + ABC_{in}$$

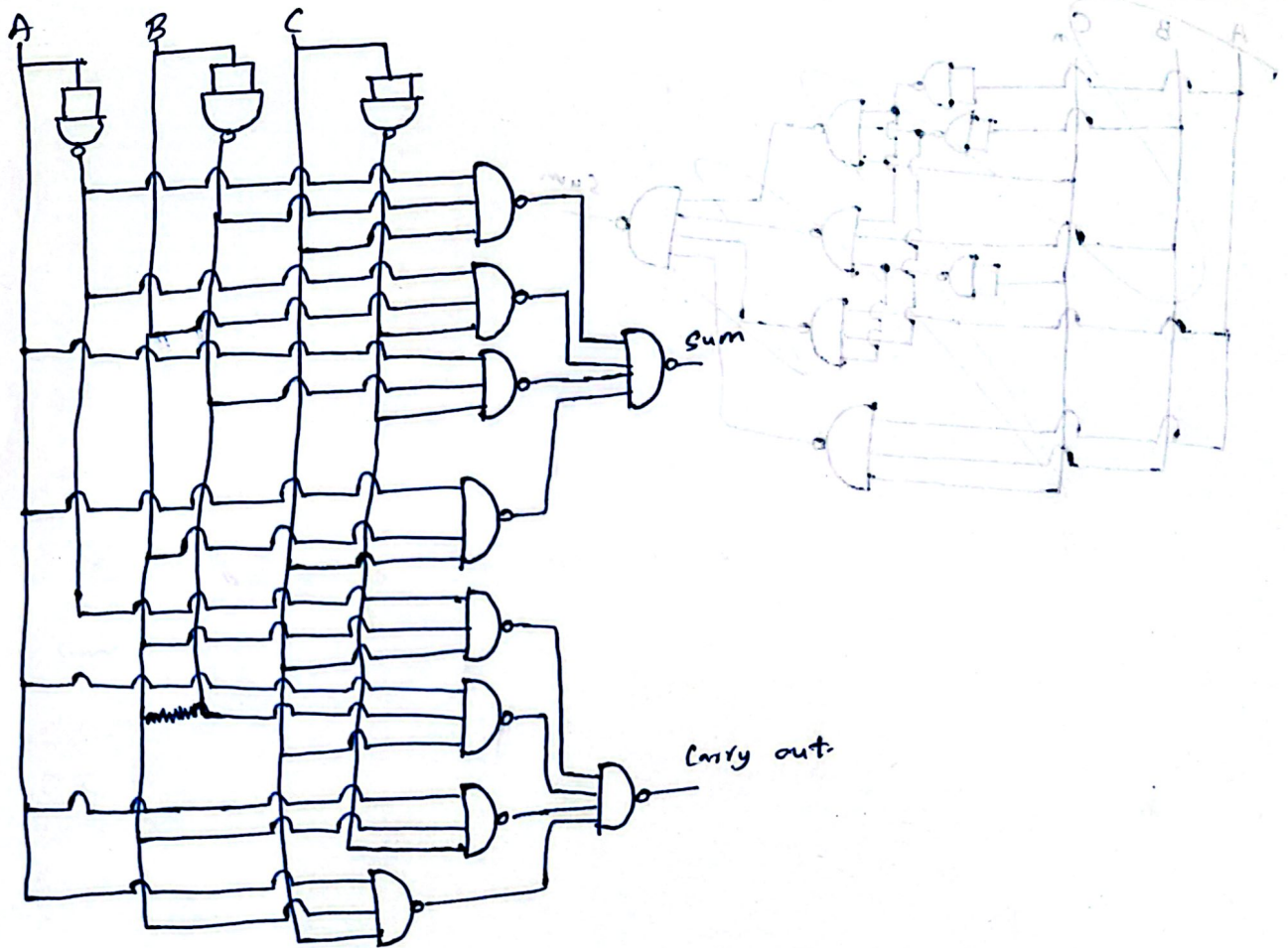
$$\text{Carry out} = \overline{A}BC_{in} + A\overline{B}C_{in} + A\overline{B}\overline{C}_{in} + ABC_{in}$$

$$\overline{A}\overline{B}C_{in} + \overline{A}B\overline{C}_{in} + A\overline{B}\overline{C}_{in} + ABC_{in}$$

$$\overline{A}\overline{B}C_{in} \cdot \overline{A}B\overline{C}_{in} \cdot A\overline{B}\overline{C}_{in} \cdot ABC_{in}$$

$$\overline{A}\overline{B}C_{in} + \overline{A}B\overline{C}_{in} + A\overline{B}\overline{C}_{in} + ABC_{in}$$

$$\overline{A}\overline{B}C_{in} \cdot \overline{A}B\overline{C}_{in} \cdot A\overline{B}\overline{C}_{in} \cdot ABC_{in}$$



③

A	B	Diff	Borrow in
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

$$\text{Diff} = \bar{A}B + A\bar{B}$$

$$= \overline{\overline{\bar{A}B + A\bar{B}}}$$

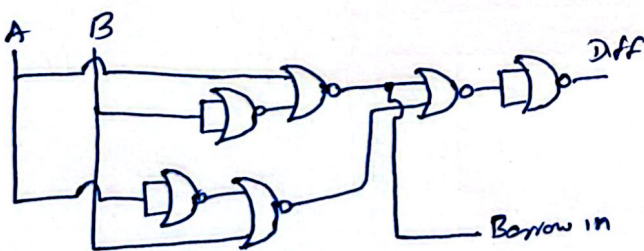
$$= \overline{\overline{\bar{A}B} \cdot \overline{A\bar{B}}}$$

$$= \overline{(\bar{A} + \bar{B}) \cdot (\bar{A} + \bar{B})}$$

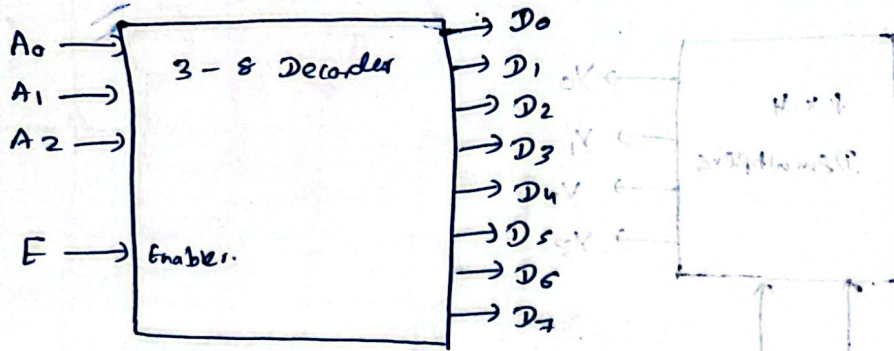
$$= \overline{(\bar{A} + \bar{B})} + \overline{(\bar{A} + \bar{B})}$$

$$\text{Borrow in} = \bar{A}B$$

$$= \overline{(\bar{A} + \bar{B})} \quad [\text{from above}]$$



4)



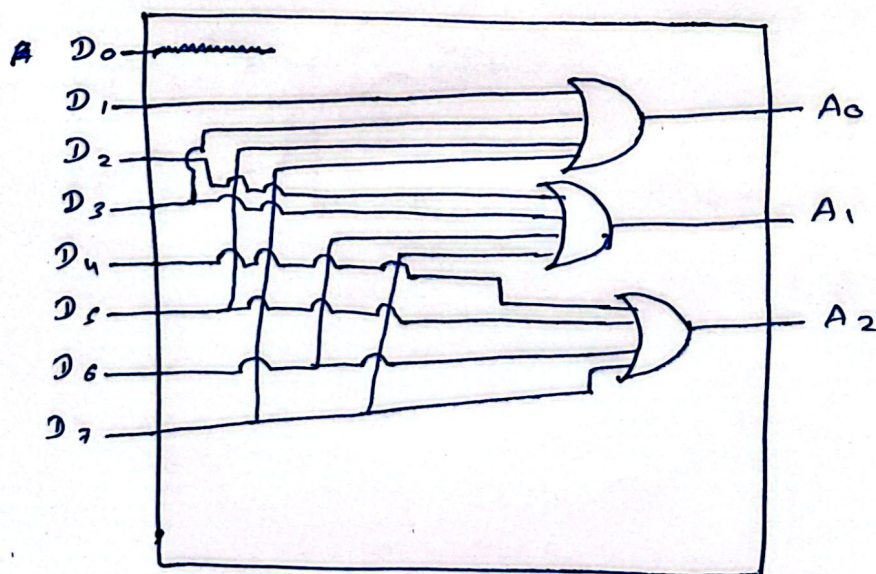
5)

Input								output		
D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0	A_2	A_1	A_0
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	0	0	0	1
0	0	0	0	0	1	0	0	0	1	0
0	0	0	0	1	0	0	0	0	1	1
0	0	0	1	0	0	0	0	1	0	0
0	0	1	0	0	0	0	0	1	0	1
0	1	0	0	0	0	0	0	1	1	0
1	0	0	0	0	0	0	0	1	1	1

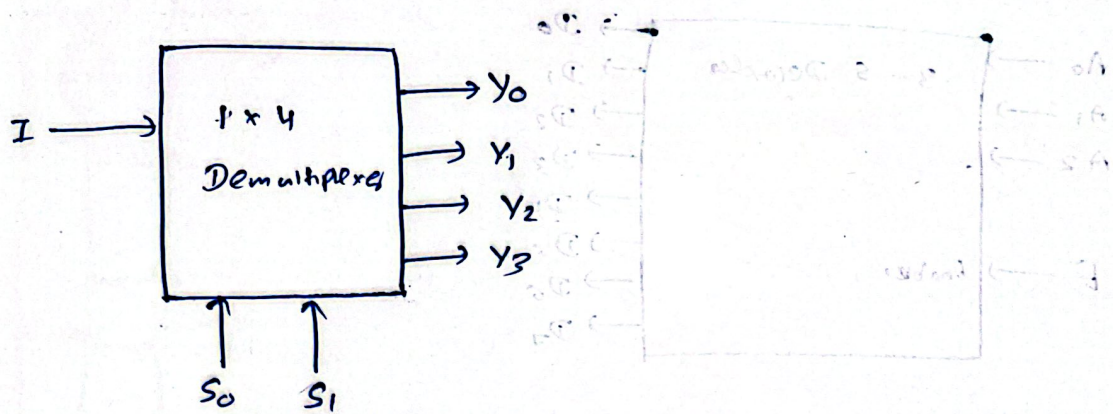
$$A_0 = D_1 + D_3 + D_5 + D_7$$

$$A_1 = D_2 + D_3 + D_6 + D_7$$

$$A_2 = D_4 + D_5 + D_6 + D_7$$



6



Inputs			Outputs									
s_1	s_0	I	Y_0	Y_1	Y_2	Y_3	Y_0	Y_1	Y_2	Y_3	Y_0	Y_1
0	0	0	1	0	0	0	0	0	0	0	0	0
1	0	0	0	1	0	0	0	0	0	0	0	0
0	1	0	0	0	1	0	0	0	0	0	0	0
1	1	0	0	0	0	1	0	0	0	0	0	0
0	0	1	0	0	0	0	1	0	0	0	0	0
1	0	1	0	0	0	0	0	1	0	0	0	0
0	1	1	0	0	0	0	0	0	1	0	0	0
1	1	1	0	0	0	0	0	0	0	1	0	0

$$\begin{aligned}
 Y_0 &= \bar{s}_1 \bar{s}_0 I \\
 Y_1 &= \bar{s}_1 s_0 I \\
 Y_2 &= s_1 \bar{s}_0 I \\
 Y_3 &= s_1 s_0 I
 \end{aligned}$$

